Estonian-Finnish Olympiad 2015

Editors: S. Ainsaar, M. Heidelberg

1. ANNIHILATION (6 points) *(by L. Franti)* An electron with kinetic energy 1 MeV travels along the z-axis and collides with a positron at rest. The particles annihilate producing a pair of two photons A and B with equal energies.

i) *(1.5 points)* Find the speed of the electron v_e in the lab frame.

ii) *(0.5 points)* Find the energy E_γ of photon A in the lab frame.

iii) *(1 point)* Find the momentum p_γ of photon A in the lab frame.

iv) *(2 points)* Find the angle α between the z-axis and the momentum of photon A in the lab frame.

v) *(1 point)* Why is it not possible for a collision process to produce just one single photon?

The rest mass of electron $m_e = 511\,\mathrm{keV}/c^2$. You can express your results in electron volts and related units.

2. HOLOGRAPHIC LENS (7 points) *(by J. Kalda)* Monochromatic light can be focused into a point using an holographic lens (better known as a *Fresnel zone plate*). This is a thin film which has a system of opaque and transparent concentric rings: the opaque rings obstruct those light waves which would arrive at the focus in the opposite phase (as compared with the light from the transparent rings). In what follows, let us assume that the diameter and focal length of such a lens $d = f = 10\,\mathrm{cm}$; the wavelength of the monochromatic incident light radiation is $\lambda = 5 \times 10^{-7}\,\mathrm{m}$.

i) *(1.5 points)* Starting enumeration from the (opaque) centre of the lens, what is the radius of the m-th transparent ring (more precisely, the circle at the middle of the ring)?

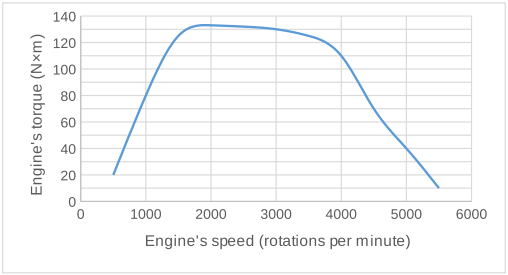
ii) *(1.5 points)* Let us consider also a glass lens of the same focal length f and diameter d, such that a parallel incident beam is collected perfectly into the focus (has aspheric surfaces). What is the minimal possible thickness of such a lens if the coefficient of refraction of the glass is $n = 1.5$?

iii) *(2 points)* If a short light pulse falls onto a lens, the behaviour of the holographic lens is very different from that of the glass lens. Indeed, if we ignore the dispersion of the glass (the dependence of n on the wavelength), the pulse duration at the focus of the glass lens is the same as that of the incident beam. Meanwhile, in the case of a short pulse ($\tau = 3 \times 10^{-14}\,\mathrm{s}$), the pulse duration at the focus of the holographic lens becomes significantly dilated. Sketch qualitatively the light intensity I at the focus of the holographic lens as a function of time. A scale is required for time t, but not for I. The speed of light $c = 3 \times 10^8\,\mathrm{m/s}$.

iv) *(1 point)* Even if the light source is an ideal laser providing a perfectly monochromatic radiation, the short pulse is no longer strictly monochromatic. Estimate the width of the range of wavelengths in the pulse of duration τ.

v) *(1 point)* Estimate the duration of this light pulse in the focus of the glass lens. Assume that in the glass, the light wave group velocity v_g depends on the wavelength at the rate $\frac{\mathrm{d}v_g}{\mathrm{d}\lambda} = 0.02 \times \frac{v_g}{\lambda}$ and that for the wavelength λ, group speed is equal to the phase speed.

3. GEAR SHIFT (3 points) *(by K.A. Saar)* The maximum torque of a car's engine depends on the engine's rotation speed (see the following figure, a larger copy is on an extra sheet).



The rotations of the engine are transferred to the wheels through a transmission gearbox. When the car is in the first gear, then the gear ratio from the engine to the wheels is 14:1; in the second gear, the gear ratio is 7:1.

i) *(2 points)* At what speed of the car should the transmission be shifted from the first gear into the second, in order for the average acceleration of the car to be maximum?

ii) *(1 point)* What is the acceleration of the car immediately before and immediately after the gear change?

Neglect any drag force. The mass of the car $m = 1400\,\mathrm{kg}$, the diameter of a wheel is $d = 60\,\mathrm{cm}$.

4. STAR WARS (8 points) *(by J. Kalda)* Here we'll study, in which region of space targets can be hit with an interplanetary ballistic missile. The launching point P is motionless in the inertial frame of reference of a gravitationally attracting star centred at S. The launching speed of the missile is fixed throughout this problem and equal to v_0, the same applies to the distance to launching point $|SP| = R$. The mass of the star is M, and the gravity constant is G. *Hint:* the total energy of a missile of mass m moving on an elliptical orbit is $E = -GMm/(2a)$, where a is the longer semiaxis of its orbit.

i) *(2 points)* The missile is launched as described above in an arbitrary direction. What is the period of the missile on its elliptic orbit?

ii) *(2 points)* According to the Kepler's I law, the missile's orbit is an ellipse, with one of the foci being at S. The position of the other focus F depends on the launching angle. While the launching angle is changed, the focus F moves along a certain curve; what curve is it? Also, determine the geometric dimensions of that curve.

iii) *(1 point)* Let us consider such a point Q on the missile's trajectory, the distance of which from the star equals to $|SQ| = r$. Determine the distance $|QF|$.

iv) *(1 point)* The position of point Q (as defined above) depends on the launching angle of the missile, and so does the distance $|PQ|$. Among all the possible positions of the point Q, what is the largest possible distance $|PQ|$?

v) *(2 points)* For questions iii and iv, we assumed that the distance r had a fixed value. Now, if we consider r as a free parameter, the position of the point Q is a function of both r and of the launching angle. What is the boundary of that region of space to which the point Q can belong?

5. RADIATOR (8 points) *(by M. Heidelberg)* Measure the heat transfer coefficient h between the environment and an aluminium profile. $h = \frac{p}{T-T_0}$, where p is the heat flux from the aluminium profile to air per unit of length; T and T_0 are the temperatures of aluminium and the environment, respectively. For measurements, keep the aluminium profile flat on your table surface, to limit nonlinear effects from turbulence. Thermal conductivity of aluminium is $k = 205\,\mathrm{W/(K\cdot m)}$ and the cross-section of the profile is $A = 36\,\mathrm{mm}^2$.

Hint: to describe the stationary temperature distribution along the profile we can derive a one-dimensional Helmholtz equation $T''(x) = \frac{h(T(x)-T_0)}{kA}$, the general solution of which is

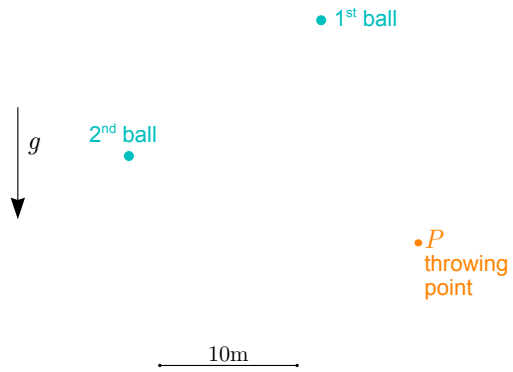
$$T(x) = T_0 + C_1 e^{x\sqrt{\frac{h}{kA}}} + C_2 e^{-x\sqrt{\frac{h}{kA}}},$$

where C_1 and C_2 are integration constants.

Equipment: Aluminium profile, wire (for heating), DC power source, infrared thermometer, ruler.

Keep the temperature of the wire below 150 °C as both the wire insulation and the table surface will start to emit smoke after that point!

6. TWO BALLS (5 points) *(by J. Kalda)* The following snapshot (a larger version is on an extra sheet) depicts two balls that were thrown simultaneously and with the same initial speed, but in different directions from point P. What was the initial speed? Use $g = 9.8\,\mathrm{m/s}^2$.



7. BOUNCY BALL (5 points) *(by J. Toots)* A homogeneous elastic ball with radius R bounces from a vertical wall in such a way that it rebounds back exactly in the same direction it came from. The ball's speed before the bounce is v at an angle α to the vertical; the ball also has an angular speed ω. There was no slipping at the contact point, but do not assume that $\omega R = v\cos\alpha$. The bounce was completely elastic, ie no kinetic energy was lost; also, the horizontal components of incident and rebound velocities have the same magnitude. The moment of inertia for a homogeneous ball is $I = \frac{2}{5}mR^2$.

i) *(1 point)* Find the angular speed ω_2 of the ball after the bounce.

ii) *(2 points)* Find the initial angular speed ω in terms of v, α and R.

iii) *(2 points)* What is the minimum value for coefficient of friction μ that such bounce could happen?

8. ELECTRIC FIELD (6 points) *(by L. Franti)* A conducting ring with radius R carries a large current I. The ring lies stationary in the xy-plane with its centerpoint at (0,0,0). An observer approaches the ring parallel to the z-axis with velocity v and measures the electric field at distance r from the axis. Assume that $v \ll c$ and $r \ll R$.

i) *(2 points)* Find the direction and magnitude of the magnetic field $B(z)$ on z axis in stationary reference frame.

ii) *(4 points)* Estimate the magnitude and direction of electric field $E(z,r)$ measured by the observer when he is at a distance z from the ring's plane.

9. SOLENOIDS (8 points) *(by S. Ainsaar).* A tightly-wound rigid solenoid coil has been partly inserted into another. They are connected to a constant current source so that the same current I flows through them; both generate a magnetic field in the same direction. The solenoids have N turns both, their length is l, their cross-sectional areas are A_1 and A_2. It may be assumed that $A_1, A_2 \ll l^2$. You may find it useful to know that the magnetic field at the centre of one solenoid taken separately is $B = \mu_0 IN/l$, where μ_0 is the permeability of vacuum.

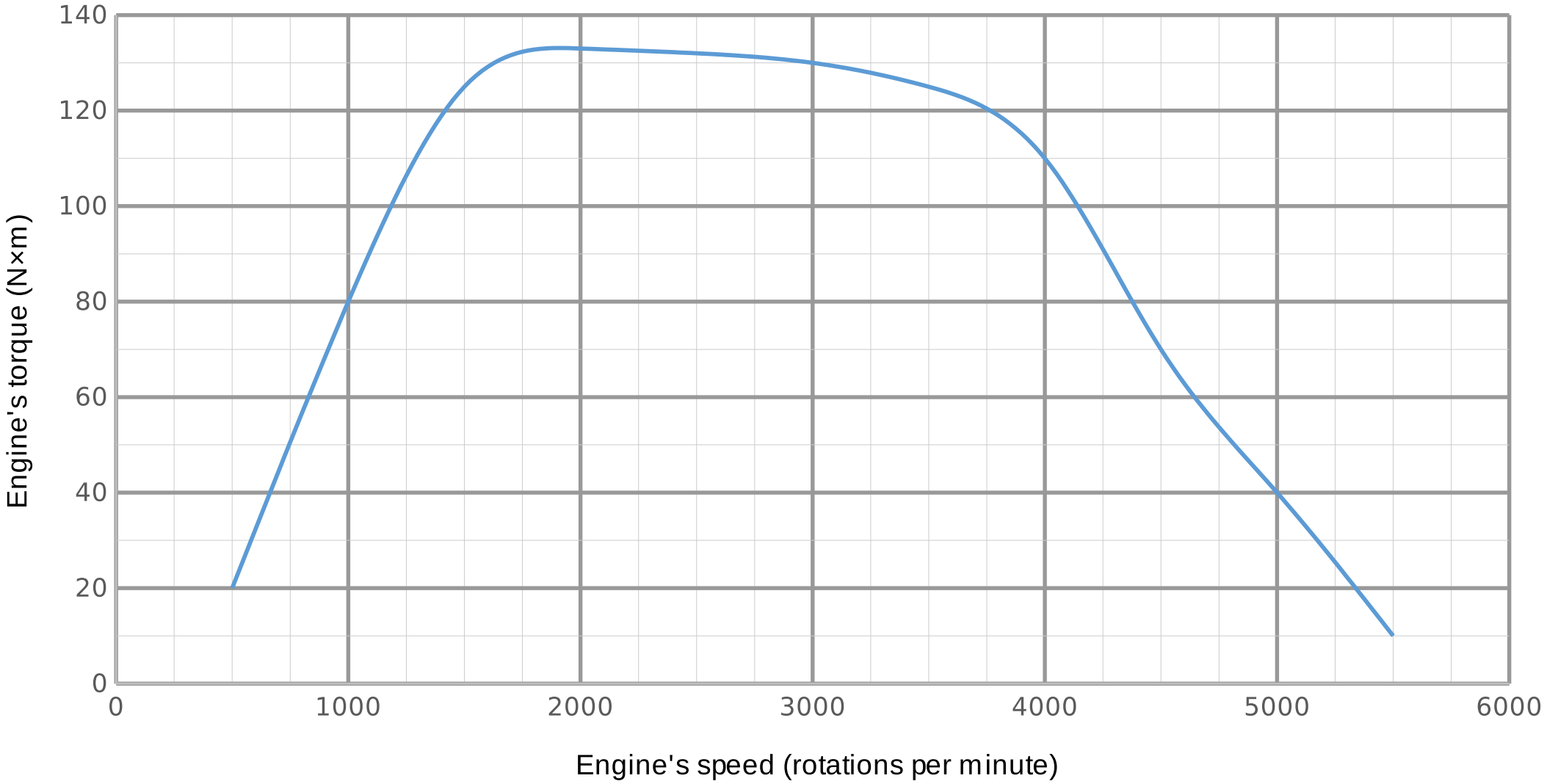
i) *(2 points)* The centres of the coils are at a distance $x < l$ along their common axis $[A_1, A_2 \ll (x-l)^2, x^2]$. Find the total energy E_m of the magnetic field in the system.

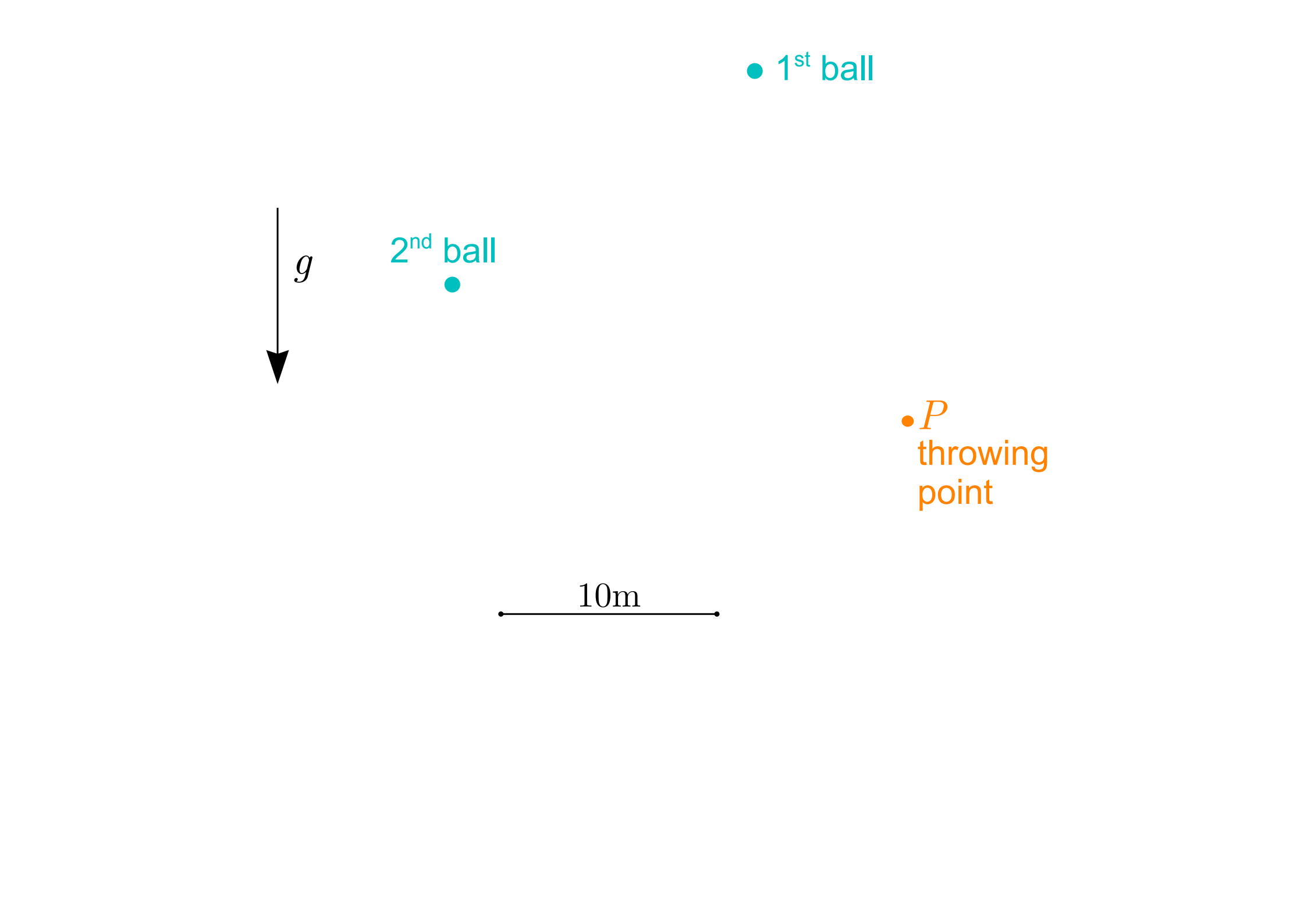
ii) *(4 points)* Find the electromotive forces $\mathscr{E}_1$ and $\mathscr{E}_2$ generated on the coils when one is pulled out with a speed v.

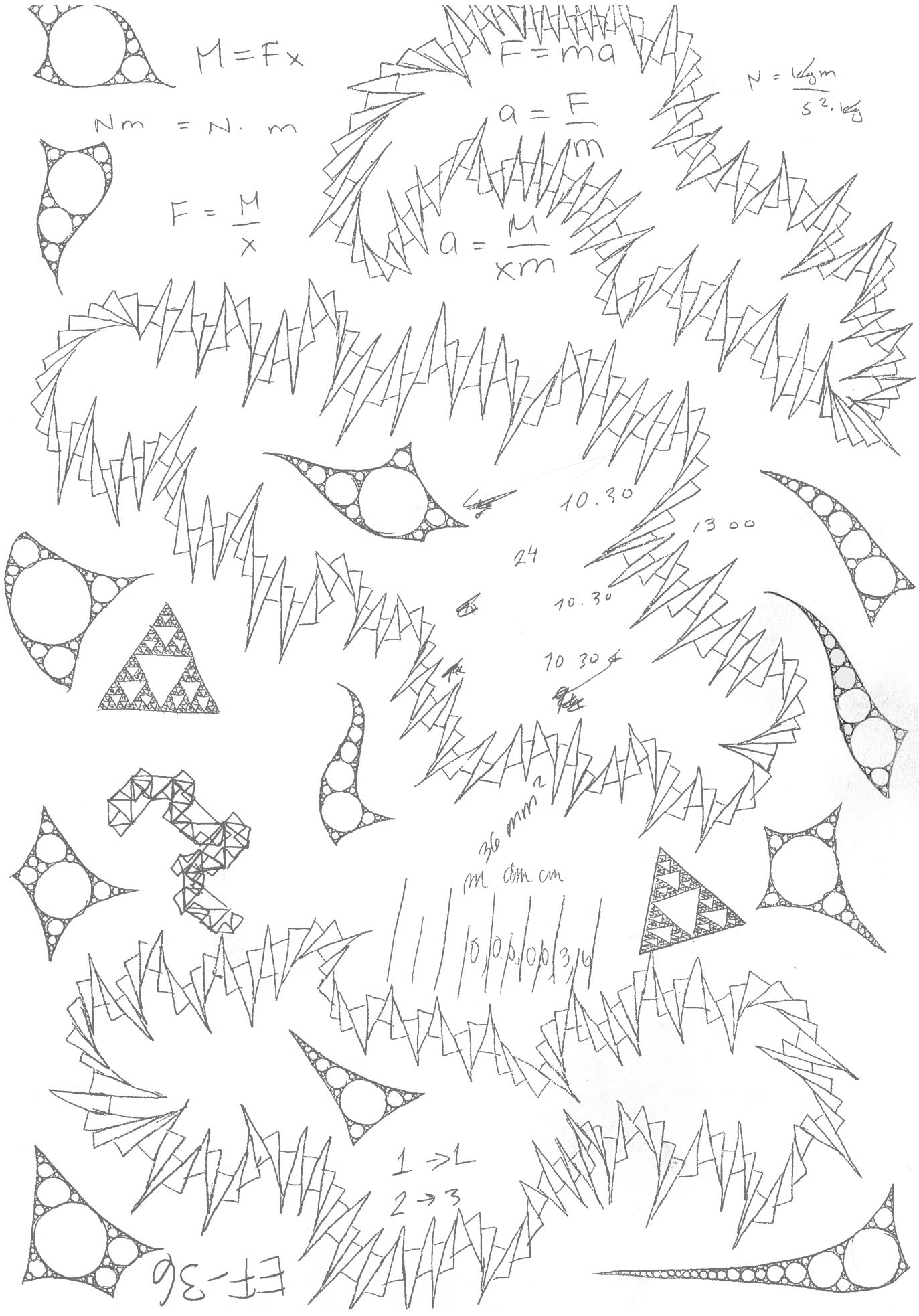
iii) *(2 points)* Find the force F needed to pull one coil outwards.

10. VAPOUR PRESSURE (8 points) *(by J. Kalda, M. Heidelberg, E. Uustalu)* Measure the room temperature saturated vapour pressure of the unknown fluid in the syringe. The volume of the bottle is $V = 0.50\,\ell$, the inner diameter of the pipe is $d = 6.0\,\mathrm{mm}$. Current air pressure p_0 and room temperature T_0 are written on the blackboard. There is no need to repeat the measurement after a successful attempt as getting the bottle free from vapours is time consuming. Estimate the uncertainty of the result.

Equipment: syringe with an unknown fluid, syringe with water, bottle, flexible pipe, ruler, plug, stand, tape.


140
120
100
80
60
40
20
0
Engine's torque (N×m)
0
1000
2000
3000
4000
5000
6000
Engine's speed (rotations per minute)


1st ball
2nd ball
g
P
throwing
point
10m


M = Fx
Nm = N · m
F = M/x
F = ma
a = F/m
a = M/xm
N = kg m/s² kg
10.30
13 00
24
10.30
10 30
36 mm²
mm dm cm
1 | | | 0,0,0,0,0,3,6
1 → L
2 → 3
ET-36

Estonian-Finnish Olympiad 2015 Solutions

1. ANNIHILATION

i) Electron's total energy $E = \gamma m_e c^2 \implies \gamma = \frac{E}{m_e c^2}$, where $E = T + m_e c^2$ (here, $T = 1\,\text{MeV}$ is the given kinetic energy). Now, $\gamma = 1/\sqrt{1 - v_e^2/c^2} \implies v_e = c\sqrt{1 - 1/\gamma^2} = c\sqrt{1 - m_e^2 c^4/E^2} = \frac{c\sqrt{T^2 + 2Tm_e c^2}}{T + m_e c^2}$. Numerically, $v_e \approx 0.941c$.

ii) The photons fly away symmetrically with respect to the electron's trajectory: in the zero-total-momentum frame, momentum conservation implies that the photons have equal momenta and, thus, equal energies, and fly in exactly opposite directions; their energies can be equal also in the positron's frame only if they are flying totally symmetrically. Each photon gets a half of the total energy in the system: $E_\gamma = \frac{1}{2}(T + 2m_e c^2) \approx 1.01\,\text{MeV}$.

iii) $E_\gamma = p_\gamma c \implies p_\gamma = E_\gamma/c = 1.01\,\text{MeV}/c$.

iv) From $E = p_e^2 c^2 + m_e^2 c^4$, the electron's momentum is $p_e = \sqrt{\frac{E^2}{c^2} - m_e^2 c^2} = \frac{1}{c}\sqrt{T^2 + 2Tm_e c^2}$. (Equivalent result can be derived from $p_e = \gamma m_e v_e$.) Momentum conservation in the z-direction then implies that $p_e = 2p_\gamma \cos\alpha \implies \alpha = \arccos\frac{p_e}{2p_\gamma} = \arccos\frac{1}{\sqrt{1 + 2m_e c^2/T}} = \arctan\sqrt{\frac{2m_e c^2}{T}} \approx 45.3°$.

v) In the center-of-mass frame, the total momentum of any system is zero. This means that if the outcome of a collision is only a single particle, then the particle's momentum must be zero in the center-of-mass frame. However, a photon's momentum can never be zero, because otherwise it would have zero energy and an infinite wavelength.

2. HOLOGRAPHIC LENS

i) Let $N = 0, 1, \ldots$ number the zones (both opaque and transparent). The optical path difference between two neighbouring zones must be $\lambda/2$ (opposite phase is demanded). The path difference between the Nth zone and the 0th zone, on the other hand, is $\Delta_N = \sqrt{r_N^2 + f^2} - f$. Therefore, $\frac{N\lambda}{2} = \sqrt{r_N^2 + f^2} - f$ and $r_N = \sqrt{\left(\frac{N\lambda}{2}\right)^2 + N\lambda f}$. Only odd-numbered zones are transparent, thus we need $r_{2m+1} = \sqrt{\left(m + \frac{1}{2}\right)^2 \lambda^2 + (2m+1)\lambda f}$.

ii) A perfectly focussing glass lens is such that all the possible light rays that go to the focus have an equal optical path length. The optical path length inside a refracting medium is n times longer than the corresponding geometric length (the phase velocity is slowed down by a factor of n). Denote the sought-after thickness by x. Equate the optical path lengths of a ray through the edge of the lens and of a ray through its centre: $\sqrt{\left(\frac{d}{2}\right)^2 + f^2} = f - x + nx \implies x = \frac{1}{n-1}\left[\sqrt{\left(\frac{d}{2}\right)^2 + f^2} - f\right] \approx 2.4\,\text{cm}$.

iii) Firstly, note that the given pulse is short enough that the whole lens never illuminates the focus — the pulse is only $\frac{\tau c}{\lambda} = 18$ periods long, but $r_{2\times 18} \approx 1.3\,\text{mm} \ll 5\,\text{cm}$. This implies that only a thin strip of the lens is illuminating the focus at a time. The intensity, when the Nth period is being observed, is proportional to the area of the Nth zone. This is $A_N = \pi\left(r_{N+1}^2 - r_N^2\right) = \pi\left(\frac{N\lambda^2}{2} + \frac{\lambda^2}{4} + \lambda f\right)$.

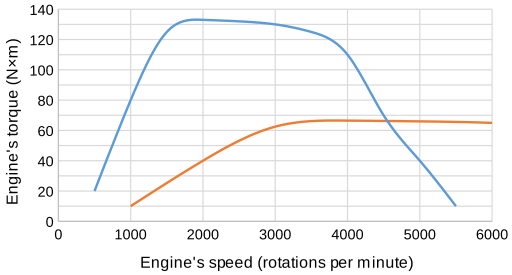
As N is proportional to time (the period of the wave is constant), the intensity will also grow linearly in time. The linear part starts at $N = 1$ with a jump and ends at $N_{\max}\frac{\lambda}{2} = \sqrt{\left(\frac{d}{2}\right)^2 + f^2} - f$ with a jump back into darkness, when the light from the edge of the lens arrives. The total duration of illumination is (approximately) $\tau_{\text{hol}} = N_{\max}\frac{\lambda}{c} = \frac{2}{c}\left[\sqrt{\left(\frac{d}{2}\right)^2 + f^2} - f\right] \approx 7.9 \times 10^{-9}\,\text{s}$.

iv) The pulse is localized into a region of space with a width $\Delta x = c\tau$. Because of the Heisenberg's uncertainty principle, the pulse is composed of photons with a range of momenta (if we take the picture that the properties of the individual photons are classical) or, from a different viewpoint, is a single photon with a somewhat uncertain momentum; either way, the characteristic width in the momentum space is $\Delta p = \frac{\hbar}{\Delta x} = \frac{\hbar}{c\tau}$. The wavelength of a photon, whose momentum p is known, is $\lambda = \frac{h}{p}$ (this is the de Broglie relation: the photon's energy is pc and also $h\nu = \frac{hc}{\lambda}$). Thus, $\Delta\lambda \approx \left|\frac{\mathrm{d}}{\mathrm{d}p}\frac{h}{p}\right|\Delta p = \frac{h\Delta p}{p^2} = \frac{h\frac{\hbar}{c\tau}}{\left(\frac{h}{\lambda}\right)^2} = \frac{\lambda^2}{2\pi c\tau} \approx 4.4 \times 10^{-9}\,\text{m}$.

v) The spread of the arrival times of waves with different wavelengths is the largest for the waves that spend the longest time inside the lens. Therefore it is enough to consider only the waves that go through the thickest part of the lens — its centre. The spread in the arrival times is $\Delta t = \Delta\frac{x}{v_g} = \frac{x\Delta v_g}{v_g^2} = \frac{x\Delta\lambda}{v_g^2}\frac{\mathrm{d}v_g}{\mathrm{d}\lambda}$. To find the group velocity v_g itself, we can use the hint (given during the examination) that in this question we may assume the group velocity and the phase velocity to be equal (in reality it would be an unusual coincidence): $v_g = v_p$ and the phase velocity $v_p = c/n$. Therefore, $\Delta t = \frac{x\Delta\lambda}{v_g^2} \times 0.02\frac{v_g}{\lambda} = 0.02\frac{xn\Delta\lambda}{c\lambda} \approx 2.1 \times 10^{-14}\,\text{s}$. The total observed pulse length behind the glass lens is $\tau_{\text{gl}} = \tau + \Delta t = 5.1 \times 10^{-14}\,\text{s}$.

3. GEAR SHIFT

i) The acceleration of the car is proportional to the torque applied to the wheels. Therefore we should keep the torque onto the wheels as big as possible, and shift gears when the torque applied onto the wheels in the second gear is bigger than the one obtained in the first gear. At the same car's speed, the engine's speed is $(14:1):(7:1) = 2$ times bigger in the first gear than in the second gear. Correspondingly, at the same engine's speed, the torque onto the wheels is two times smaller in the second gear than in the first gear. Therefore we can draw another graph where the engine's torque is twice as small and the engine's speed is two times bigger. The intersection of those graphs is the point where the gear should be changed.



From the graph we read that at that moment in the first gear, the engine's torque is $\tau_e = 66.5\,\text{N}\cdot\text{m}$ and the engine's speed is $\omega_e = 4550\,\text{rpm}$. The wheels' angular speed is then $\omega_w = \frac{1}{14}\omega_e$ and the car's speed is $v = \frac{1}{2}\omega_w d = \frac{1}{28}\omega_e d$. Let's convert $1\,\text{rpm} = 120\pi\,\text{rad/h}$ and $d = 6 \times 10^{-4}\,\text{km}$. Thus, $v \approx 37\,\text{km/h}$.

ii) At the optimum point, the acceleration before and after gear change is the same. From Newton's second law, it equals $a = F/m = \frac{2\tau_w}{md}$, where the net torque applied to the wheels is $\tau_w = 14\tau_e$. Hence, $a = \frac{28\tau_e}{md} \approx 2.2\,\text{m/s}^2$.

4. STAR WARS

i) The period of a Keplerian orbit having major semi-axis a can be expressed using relation $T^2 = \frac{4\pi^2 a^3}{GM}$. However, if we didn't know the formula, we should recall Kepler's 3rd law $T^2 \sim a^3$ and simply express the period for a circular orbit with $a = r$. Therefore, $T^2 = \frac{4\pi^2 r^2}{v^2}$, were we need to insert v from $\frac{v^2}{r} = \frac{GM}{r^2}$.

For finding the period, we can get the major semi-axis from the expression for orbit's

total energy $-\frac{GMm}{2a} = \frac{mv^2}{2} - \frac{GMm}{R}$, equivalently $a = \frac{GM}{2\frac{GM}{R} - v^2} = \frac{1}{\frac{2}{R} - \frac{v^2}{GM}}$. Finally $T = \frac{2\pi a^{3/2}}{\sqrt{GM}}$.

ii) According to the properties of an ellipse, the distances l_1 and l_2 drawn from any point on the ellipse to its two foci add up to constant value $l_1 + l_2 = 2a$. Since $l_1 = R$, the other focus must be at distance $l_2 = 2a - R$, so the other focus F lies on a circle with radius $|PF| = l_2 = 2a - R$ around P.

iii) This is the previous part in reverse. Since we have distance to one focus given $|QS| = r$, we can express the distance to the other focus $|QF| = 2a - r$.

iv) The points P, Q, and F form a triangle, for which we have expressed two of its side lengths: $|PF|$ and $|QF|$. The third side QP must satisfy $|QP| \le |PF| + |QF|$. The $|QP|_{\max}$ is obtained for a degenerated triangle (all the three points lying along a common line). Therefore, $|QP|_{\max} = 4a - R - r$.

v) It turns out that for all the maximal distance points Q the sum of $|SQ| + |PQ|$ is constant and equal to $4a - R$. That means it also defines an ellipse with focal points S and P and major semi-axis $2a - R/2$. This ellipse is the area that we can hit.

5. RADIATOR

We wind the wire around one end of the aluminium profile and supply current to it for heating. In most cases it was OK to supply up to 5A (the maximum possible) through the wire, only when it was wound very compact it could get too hot. This should give us good 100 °C to measure on the profile, but the experiment also worked if smaller currents/temperatures were used. Now we wait until the temperature in the profile stabilises, this takes about five to ten minutes and we can check if it has stabilised with the thermometer.

When the temperature has stabilised we measure and write down the temperature values along the part of the profile that is not covered with wire using a reasonable distance interval. We can see that that the other end of the profile is still at room temperature (the difference is below or near the resolution of this thermometer). This means that in the given solution for Helmholtz equation,

$$T(x) = T_0 + C_1 e^{x\sqrt{\frac{h}{kA}}} + C_2 e^{-x\sqrt{\frac{h}{kA}}},$$

the integration constant $C_1 = 0$.

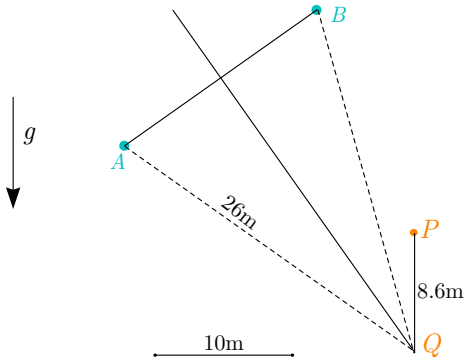
Now one way is to use $T(0)$ to express $C_2 = T(0) - T_0$ and use any other temperature measurement to calculate h. A better way is to plot the temperature difference in a logarithmic scale: $\log(T(x) - T_0) = -\sqrt{\frac{h}{kA}}x + C_2$, and use the slope of the graph $a = -\sqrt{\frac{h}{kA}}$ to calculate h. This gives a more accurate value for h and allows us to better estimate the uncertainty by finding the range of slopes that can reasonably be drawn on the graph.

The correct answer is in the range $h = 0.3\,\mathrm{W/K\cdot m}$ to $h = 0.5\,\mathrm{W/K\cdot m}$, it depends slightly on how the profile is placed on the table. The heat transfer coefficient is higher when the profile is on the edge of the table or slightly raised due to the wiring.

6. TWO BALLS

Let us consider the process in a free-falling frame of reference; them the both balls will move with constant velocities. The speeds are equal, but directions different; hence, for any moment of time, they are at equal distance from the throwing point Q. Thus, point Q can be found as the intersection point of the perpendicular bisector to the segment AB (connecting the balls), and the vertical line drawn from the point P.

In the lab frame, the point Q, however, is a free-falling point, and by time t has travelled distance $|PQ| = gt^2/2$. Using the provided scale we find from the figure that $|PQ| \approx 8.6\,\mathrm{m}$, hence $t = \sqrt{2|PQ|/g} \approx 1.3\,\mathrm{s}$. Hence, the throwing speed $v = |AQ|/t \approx 20\,\mathrm{m/s}$; here we have used reading from the figure, $|AQ| \approx 26\,\mathrm{m}$.



7. BOUNCY BALL

i) Energy is conserved: $\frac{mv^2}{2} + \frac{I\omega^2}{2} = \frac{mv^2}{2} + \frac{I\omega_2^2}{2} \implies |\omega_2| = |\omega|$. As the ball receives some angular momentum from the wall (the net force from the wall is not directed through the ball's centre), the angular velocity changes its sign — the rotation flips its direction and $\omega_2 = -\omega$.

ii) The angular momentum is conserved with respect to the contact point, because all the forces are directed through the point. Let $a = R\cos\alpha$ be the distance from the trajectory of the center of mass to the contact point. Then the aforementioned angular momentum conservation reads $I\omega - mav = I\omega_2 + mav$. As $\omega_2 = -\omega$, this gives $\omega = \frac{mav}{I} = \frac{5v\cos\alpha}{2R}$.

iii) At the verge of slipping on a surface, the angle between the surface normal and the resultant of the reaction force and the frtiction force equals $\arctan\mu$. Here this angle is $\frac{\pi}{2} - \alpha$, thence $\mu \ge \cot\alpha$.

8. ELECTRIC FIELD

i) By the Biot-Savart law, every piece of the ring with length and direction $\mathrm{d}\vec{l}$ creates a magnetic field $\mathrm{d}\vec{B} = \frac{\mu_0 I\,\mathrm{d}\vec{l}\times\hat{a}}{4\pi a^2}$, where $\vec{a}$ is the displacement vector from the piece to the point where we are calculating $\vec{B}$, and $\hat{a} = \vec{a}/a$. By symmetry, the magnetic field on the z-axis adds up to being exactly along the z-axis, because the perpendicular components cancel out. $\mathrm{d}B_z$ is proportional to $\mathrm{d}l$ and $\mathrm{d}\vec{l} \perp \hat{a}$. The vector $\mathrm{d}\vec{l}\times\hat{a}$ is at an angle $\alpha = \arctan\frac{r}{z}$ from the horizontal. Therefore we can add up the contributions and write $B(z) = \frac{\mu_0 I\times 2\pi R\sin\alpha}{4\pi(z^2+R^2)} = \frac{\mu_0 IR^2}{2(z^2+R^2)^{3/2}}$. The field is along the z-axis.

ii) The electric field measured by a charged observer must be in the same direction as the Lorentz force acting on the observer in the laboratory frame of reference. Therefore the electric fieldlines are circles around the z-axis. Now we may apply Faraday's induction law to such a circle, as it approaches the ring: the total electromotive force $E(z,r)\times 2\pi r = \Phi = -\frac{\mathrm{d}\Phi}{\mathrm{d}z}v \approx -\frac{\mathrm{d}}{\mathrm{d}z}B(z)\times\pi r^2\times v \implies E(z,r) = -\frac{rv}{2}B'(z) = \frac{3\mu_0 IrR^2vz}{4(z^2+R^2)^{5/2}}$.

9. SOLENOIDS

i) The overlapping region has magnetic field $2B$, rest of the coils' insides has B. Let the inner coil have an area of A_2. Energy density, where the magnetic field is B, is $\frac{B^2}{2\mu_0}$. Therefore $E_m = \frac{B^2}{2\mu_0}(A_1 l - A_2(l-x) + A_2 x) + \frac{(2B)^2}{2\mu_0}A_2(l-x) = \frac{B^2}{2\mu_0}[A_1 l + A_2(3l-2x)] = \frac{\mu_0 I^2 N^2}{2l^2}[A_1 l + A_2(3l-2x)]$.

ii) The outer coil is intersected by all the internal coil's magnetic flux BA_2. After time dt this flux is enclosed by fewer turns of the outer coil — by the $Nv\,dt/l$ turns that are on the length $v\,dt$. Thus the flux enclosed by the outer coil changes with a rate $\mathscr{E}_1 = \dot{\Phi}_1 = BA_2Nv/l = \mu_0 A_2 IN^2 v/l^2$. The inner coil is also intersected by the flux BA_2, thus $\mathscr{E}_2 = \mathscr{E}_1$; those electromotive forces are in the same direction.

iii) The work done when pulling a coil out is balanced in two places: the magnetic field

energy decreases and our constant current source dissipates some energy (because our electomotive forces work against it). In total, the applied mechanical power $Fv = \frac{\mathrm{d}E_m}{\mathrm{d}x}v + I(\mathscr{E}_1 + \mathscr{E}_2)$, from where $F = -\frac{B^2A_2}{\mu_0} + \frac{2IBA_2N}{l} = \frac{\mu_0 I^2N^2A_2}{l^2}$.

10. VAPOUR PRESSURE

We construct a manometer by fixing the pipe into a "U" shape using the stand, and filling the tube partially with water. With a small trouble we should get the water to the bottom of "U" shape and then we can measure the pressure difference from differences of depth $\Delta p = \rho g \Delta h$. After we attach a bottle to one end, we should let the pressures equalise through a needle hole in the bottle. We then squirt the unknown liquid to the bottle and close the needle hole with tape as fast as possible. We then shake the bottle to hasten the vaporisation and write down the difference of depths Δh after it has reached a stable value. During all this we should be careful to heat the bottle with our body as little as possible.

If the volume would have been fixed we would get the vapour pressure directly from the manometer reading as according to Dalton's law. But since the diameter of the pipe was not that small we should take the relative volume increase to account.

If the depth difference is Δh, we have a relative increase of pressure $n_p = \frac{\rho g \Delta h}{p_0}$ and a relative increase of volume $n_V = \frac{S\Delta h}{2V_0}$. Assuming an ideal gas, this gives us relative molar increase $n_n = \frac{(1+n_p)(1+n_V)p_0V_0 - p_0V_0}{p_0V_0} \approx n_p + n_V$. Substituting in the given values we get $n_n \approx 0.0987\,\mathrm{m}^{-1} \times h + 0.00471\,\mathrm{m}^{-1} \times h \approx 0.1034\,\mathrm{m}^{-1} \times h$.

Since all the molar increase is due to the vapour, we can get the vapour pressure (partial pressure exerted by vapour) by multiplying the pressure with the molar fraction of vapour $p_x = (p_0 + \Delta p)\frac{n_n}{1+n_n}$.

The unknown liquid was ethanol with vapour pressure of $p_x = 6.52\,\mathrm{kPa}$ at $21.6\,^\circ\mathrm{C}$. However, the grading scheme was not insistent on the exact value as it proved technically quite difficult to get it correct.